10-DAY • HANDS-ON • BEGINNER-FRIENDLY

AI Skills Lab: Learn, Create & Automate

10-Day Syllabus



Instructor-led course • October 2026

Course overview & audience

A practical lab for beginners who want to understand AI, build useful projects, and automate everyday workflows.

10

students

A focused cohort

20h

total time

10 × 2-hour sessions

Hands-on

format

Instructor-led practice

LEARN FOUNDATIONS → CREATE PROJECTS → AUTOMATE WORKFLOWS

By the end, students can...

01

Explain AI foundations and responsible use

02

Engineer prompts that reliably perform

03

Build no-code workflows and API integrations

04

Design and demo a capstone project

05

Review, reflect, and improve together

10 days. Three deliberate phases.

DAYS 1–4

Foundations

AI basics • prompts • productivity • research

DAYS 5–8

Assistants & Automation

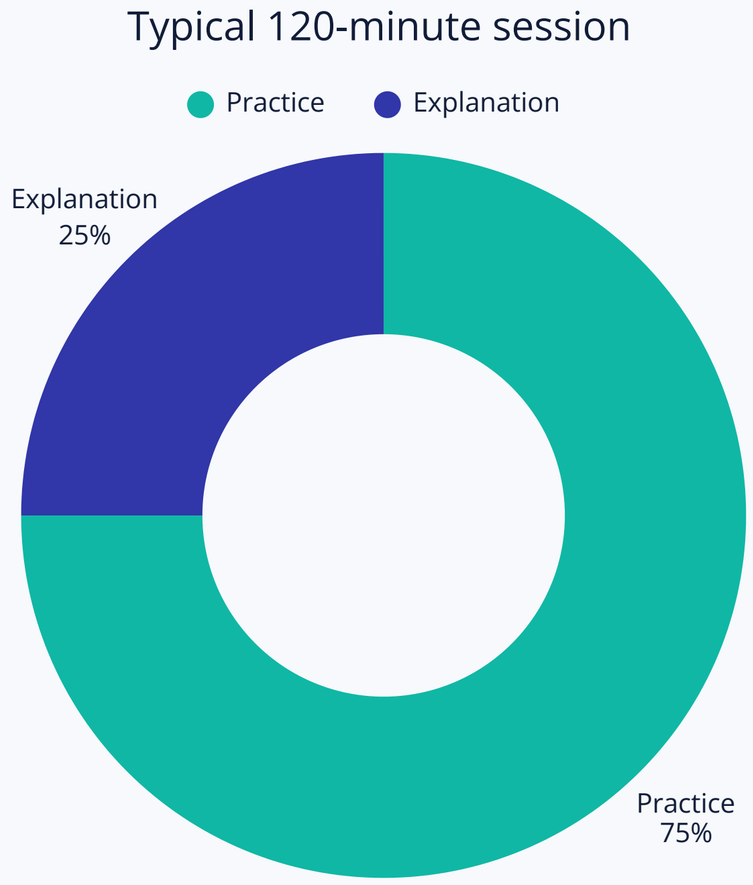
Chatbots • workflows • integration • APIs

Capstone

Design • build • test • demo

KNOW → MAKE → SHIP

Practice is the main event



70–80%

hands-on practice

1. Recap / intro
2. Teaching
3. Demo
4. Build
5. Test / peer review
6. Exit ticket

Responsible AI is a daily practice

Privacy

Keep sensitive information out.

Hallucinations

Recognize errors; check claims.

Bias

Look for skewed assumptions.

Verification

Validate before using outputs.

Human judgment

Stay critical and accountable.

Safe data

Prefer samples and anonymization.

Day 1 — AI Foundations

LEARNING OBJECTIVES

Understand core terms, applications, limits
Recognize ethical considerations

120-MINUTE AGENDA

- 10m Recap / intro
- 20m Foundations teaching
- 20m Guided tool demo
- 40m Prompt experiments
- 20m Peer review

HANDS-ON LAB

Experiment with basic prompts in an approved assistant.

DELIVERABLE

AI capabilities summary

et

HOMEWORK

Reflect on AI in daily life

Day 2 — Prompt Engineering

LEARNING OBJECTIVES

Use role, context, task, constraints, format
Refine prompts through iteration

120-MINUTE AGENDA

- 10m Recap
- 25m Prompt concepts
- 20m Iteration demo
- 40m Create + refine
- 15m Peer review

HANDS-ON LAB

Build multiple prompt versions for one defined output.

DELIVERABLE

Best prompt + comparison notes

et

HOMEWORK

Prompts for a personal project

Day 3 — AI for Learning & Productivity

LEARNING OBJECTIVES

Apply AI to study, writing, time management
Integrate AI into practical workflows

120-MINUTE AGENDA

- 10m Recap
- 20m Productivity teaching
- 25m Tool demo
- 40m Build an aid
- 15m Peer review

HANDS-ON LAB

Create an AI-assisted study or productivity aid.

DELIVERABLE

Functional prompt or workflow

et

HOMEWORK

Test it in daily productivity

Day 4 — Research, Credibility & Creative Tools

LEARNING OBJECTIVES

Evaluate sources and verify research
Use generated content ethically

120-MINUTE AGENDA

- 10m Recap
- 25m Credibility + copyright
- 20m Creative tools demo
- 40m Validate + create
- 15m Peer review

HANDS-ON LAB

Compare AI research with credible sources; create an artifact.

DELIVERABLE

Annotated summary + creative output

et

HOMEWORK

Reflect on copyright

Day 5 — No-Code Assistants & Chatbots

LEARNING OBJECTIVES

Design clear assistant instructions
Ground responses in approved sources

120-MINUTE AGENDA

- 10m Recap
- 25m Assistant basics
- 20m Chatbot demo
- 40m Build knowledge flow
- 15m Test + review

HANDS-ON LAB

Design and test a no-code chatbot assistant.

DELIVERABLE

Functional chatbot prototype

et

HOMEWORK

Prepare chatbot test cases

Day 6 — n8n Automation Basics

LEARNING OBJECTIVES

Use triggers, nodes, and actions
Read execution history and debug

120-MINUTE AGENDA

- 10m Recap
- 25m Workflow concepts
- 20m Guided workflow demo
- 40m Build automation
- 15m Debug + review

HANDS-ON LAB

Create and test a workflow automating a simple task.

DELIVERABLE

Workflow file + explanation

et

HOMEWORK

Reflect on automation use cases

Day 7 — AI Model Integration

LEARNING OBJECTIVES

Use prompt templates + structured outputs
Validate quality and correctness

120-MINUTE AGENDA

- 10m Recap
- 25m Integration teaching
- 20m Examples demo
- 40m Build templates
- 15m Test + review

HANDS-ON LAB

Deploy structured model prompts with validation.

DELIVERABLE

Template + validation report

et

HOMEWORK

Prepare integration use cases

Day 8 — APIs, JSON & Webhooks

LEARNING OBJECTIVES

Understand requests, responses, and JSON
Use webhooks and credentials safely

120-MINUTE AGENDA

- 10m Recap
- 25m API + JSON basics
- 20m Calls + webhooks demo
- 40m Build integration
- 15m Test + review

HANDS-ON LAB

Connect AI services through APIs and webhooks securely.

DELIVERABLE

Integration workflow + credentials plan

et

HOMEWORK

Document workflow steps

Eight days → eight portfolio artifacts

DAY 01

AI capabilities summary

DAY 02

Best prompt + comparison

DAY 03

Productivity prompt / workflow

DAY 04

Research annotation + artifact

DAY 05

Chatbot prototype

DAY 06

n8n workflow file

DAY 07

Integration template

DAY 08

API workflow

Day 9 — Capstone: Design, Build & Test

LEARNING OBJECTIVES

Define a relevant problem and workflow
Build, test, and troubleshoot an integrated project

120-MINUTE AGENDA

- 10m Recap
- 20m Problem + workflow plan
- 60m Build + iterate
- 20m Peer troubleshooting
- 10m Exit ticket

HANDS-ON LAB

Develop one chosen capstone option through iterative testing.

DELIVERABLE

Demo-ready prototype

HOMEWORK

Prepare demo presentation

Day 10 — Demo, reflect, move forward.

40m Student demos
20m Course quiz
30m Group reflection + feedback
10m Wrap-up + next steps

DELIVERABLE • completed quiz + demo presentation



Choose a capstone that matters

Smart Study Assistant

Support study habits with guided, useful responses.

School Information Chatbot

Provide approved information interactively.

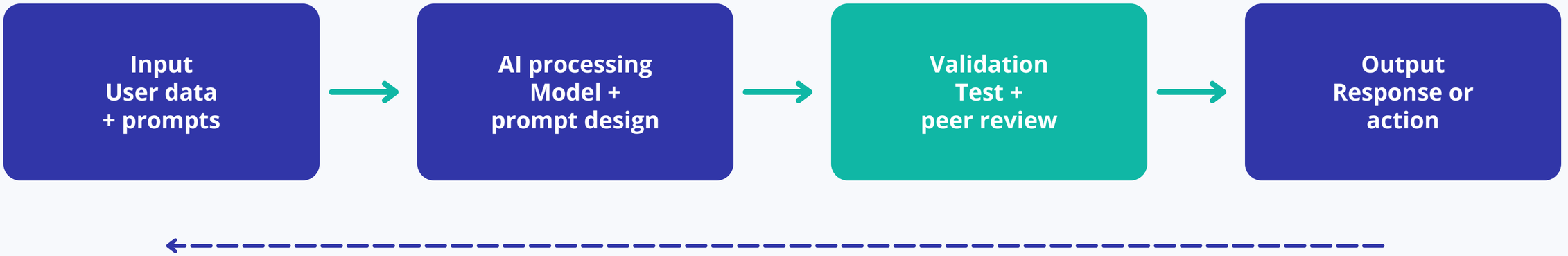
Automated Quiz Generator

Use workflow automation + AI to create quizzes.

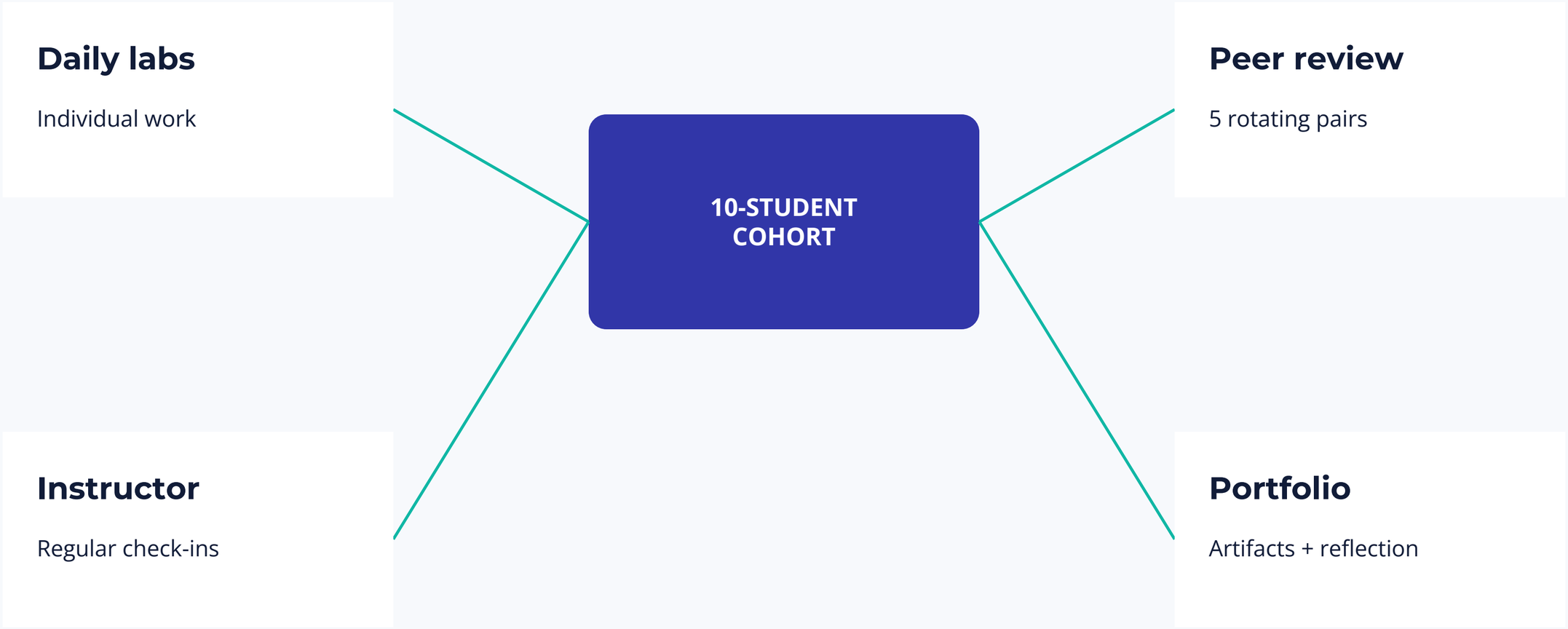
Community Problem Solver

Address a local issue with a practical AI workflow.

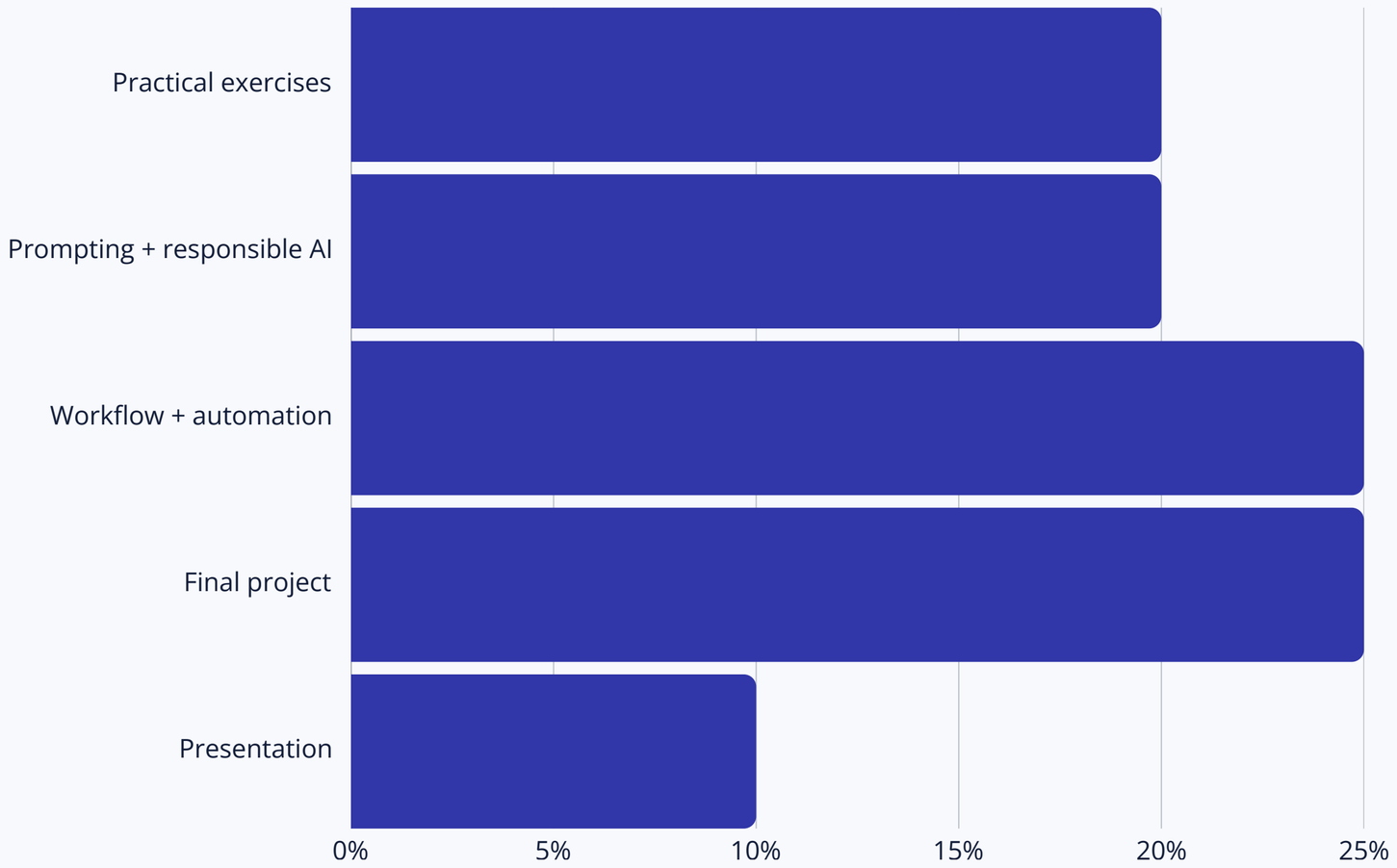
A reusable project architecture



Individual ownership. Structured support.



Assessment rewards practical mastery



25%

workflow & automation

+ 25% final project

Half the grade reflects building something that works.

Before you call it finished...

- ✓ Useful and relevant
- ✓ Functional and reliable
- ✓ Clear workflow quality
- ✓ Testing is complete
- ✓ Privacy and data safety
- ✓ Explanation + documentation
- ✓ Limits + ethics acknowledged

Set up before students arrive

1. **Approved AI assistant access**
2. **Modern browser + permissions**
3. **n8n environment ready**
4. **Model / API access if available**
5. **Non-sensitive sample datasets**
6. **Submission folder created**

Verify platform availability, account age requirements, and permissions before class.

Safe practice is non-negotiable

- ! Never expose API keys or secrets
- ! Use sample or anonymized data
- ! Validate every AI output
- ! Log errors without sensitive info
- ! Test edge cases thoroughly
- ! Treat outputs as claims—not facts

A portfolio of evidence—not promises

Prompts

Tested prompt set

Workflows

No-code automation

Capstone

Demo-ready prototype

Reflection

Learning portfolio

Expected outcomes depend on active participation; no result is guaranteed.

Finish strong. Keep building.

01

Review

all deliverables + feedback

02

Submit

your complete portfolio

03

Continue

AI practice and learning

04

Explore

advanced tools + APIs

Connect

with cohort + instructor

The lab ends. The learning loop stays open.